Computer Graphics I

4003-570 / 4005-761

Homework #2: Frame Buffers and Display Systems

Date assigned: March 14th 2013

Date due: March 21st 2013

Please submit your work in the dropbox on myCourses

Answer any three (3) of the following questions.

1. List the operating characteristics for the following display technologies: raster refresh system, vector refresh systems, plasma panels, and LCDs.

2. Determine the resolution (pixels/cm) in the x and y directions for the video monitor in use on your system. Determine the aspect ratio, and explain how relative proportions of objects can be maintained when resizing objects.

3. Consider three different raster systems with resolutions of 640 by 480, 1280 by 1024, and 2560 by 2048. Assuming 8 bits per byte, what size frame buffer (in bytes) is needed for each of these systems to store 12 bits per pixel? Assuming 8 bits per byte, what size frame buffer (in bytes) is needed for each of these systems to store 24 bits per pixel?

4. How long would it take to load a 640 by 480 frame buffer with 12 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load a 24 bit per pixel frame buffer with a resolution of 1280 by 1024 using this same transfer rate?

5. How much time is spent scanning across each row of pixels during screen refresh on a raster system with a resolution of 1280 by 1024 and a refresh rate of 60 frames per second.

6. Assuming that a certain full-color (24-bits per pixel) RGB raster system has a 512-by-512 frame buffer, how many distinct color choices (intensity levels) would we have available? How many different colors could we display at any one time?

Please show how you arrive at your answers for the questions that involve computation (partial credit will be given even if your answer is not correct).

Submit as a pdf to myCourses.